

How Many Local Epochs? Communication Cost and Variable Selection in Federated Lasso for Multi-Site Studies

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Problem. Multi-site pharmaceutical research increasingly depends on data that cannot be centralized — records behind institutional firewalls, trials under separate data-use agreements, claims databases no single sponsor controls. Federated learning shares model parameters rather than patient records. When the goal is *which variables matter* — candidate biomarkers, covariates for a safety model — the natural tool is a sparse regression such as the Lasso. Every federated protocol must choose E , the number of local update steps each site takes between aggregation rounds; larger E buys fewer communication rounds but risks client drift. For coordinate descent on the Lasso the concern is specific: because the ℓ_1 penalty zeros coordinates by soft-thresholding, drift can lead sites to prune different predictors — the damage appears as a different variable list, not merely a worse fit.

Setup. We evaluate a synchronous federated coordinate-descent Lasso using the publicly available STAT 102B high-dimensional regression dataset obtained from github.com/KeivanBolouri/federated-lasso. After pre-processing, the data contain $n = 800$ observations and $p = 600$ predictors and are computationally partitioned into $K = 3$ disjoint sites with sample sizes $m_j = 400, 240, 160$. Each site selects its regularization parameter independently. Because the true coefficient support is unknown, a centralized Lasso fitted to the pooled sample serves as the reference throughout. Each site then runs E full coordinate-descent passes per round, and the server aggregates by sample-size weight; raw data never leaves a site. We sweep $E \in \{1, 2, 3, 5, 8, 10, 15, 20\}$, recording communication rounds, upload cost, and agreement between the federated active set and that reference (197 active predictors).

Results. Recall against the reference falls monotonically from 86.3% to 69.5% as E grows from 1 to 20, while precision stays flat (42.8–46.2%). However, F1 and Jaccard are not monotone: both peak at $E = 3$ (0.583, 0.412), above their values at $E = 1$ (0.572, 0.401). Moving from $E = 1$ to $E = 3$ cuts rounds from 27 to 13 and upload from 379.7 to 182.8 KiB — a 52% reduction — while slightly improving agreement with the centralized fit, so the two objectives did not conflict in this range in this run. Beyond $E = 5$, upload cost continues to decline while F1 and Jaccard generally decrease; $E = 15$ and $E = 20$ both require six rounds and 84.4 KiB of upload, although their support-agreement metrics differ slightly. At $E = 10$ the federated model matches 137 of the 197 reference predictors against 170 at $E = 1$ — the same data yielding a materially different candidate list.

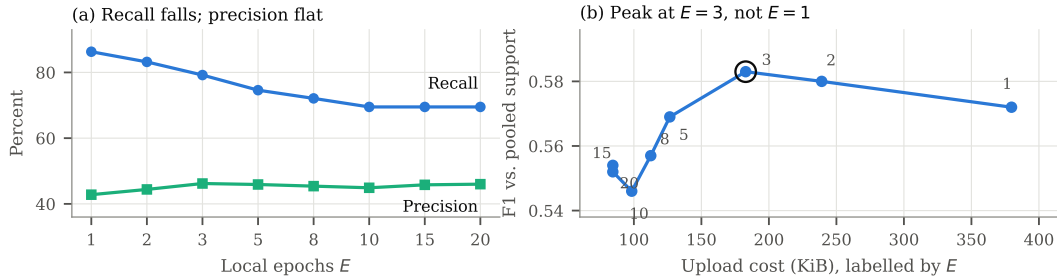


Figure 1: Recall declines monotonically in E while precision is flat (a), so agreement with the centralized fit peaks in the interior at $E = 3$ rather than at $E = 1$ (b). In this single run, reducing communication from $E = 1$ to $E = 3$ did not reduce observed F1.

Implication. A preliminary coarse grid $\{1, 5, 10\}$ favored $E = 5$ on final penalized objective (25.8165 against 25.8906 at $E = 1$), missing the $E = 2$ –3 region entirely. The objective and the selection metrics disagreed about which schedule was best, suggesting that the penalized objective alone may be insufficient for choosing E when support stability is important.

Limitations. Results come from a single draw of one dataset with no error bars, so the $E = 1$ versus $E = 3$ gap is not separated from sampling noise, and the data are not multi-site clinical data — the framing above is motivation, not demonstrated performance. Because the true support is unknown, the pooled Lasso is an estimated reference and the metrics measure agreement rather than correctness. The recall pattern is consistent with client drift, though drift was not measured directly, and no comparison is made against FedProx, SCAFFOLD, or ADMM-based distributed Lasso — replication with known true support, direct drift measurement, and those comparisons are in progress.